KSeries production database columns and constants

(from Vision3D CR4 and last Vision20 CR5)

Contents:

1 Introduction

This document presents the database used by the Vi technology inspection system to save the inspection and review results.

This document is confidential and must not be distributed without Vi technology agreement.

Modifications made on the database in this version are displayed in blue. They are also presented in §3 “What’s new in version 5.0?”.

2 Warning

Vi TECHNOLOGY recommends that customers do not develop any application that access to this database as it could impact the performances and stability of the whole system:

- applications that would overload the database server with too much or too long requests would prevent the Superviseur software to save or read inspection results in time. It can impact the inspection cycle time or worst, it can stop the production on the AOI that would reach the time-out!
- Vi TECHNOLOGY can’t be taken as responsible for such instabilities appearing on the line as soon as such requests are made in the database and slow down the server.
- Vi TECHNOLOGY will not be able to help customer for such communication issues (slowness, lost connections...) as we don’t master the overall system.
- The database format can be changed when Vi TECHNOLOGY enhances his Vision solution. As result, customers will have to adapt their own applications.

Vi TECHNOLOGY recommends that customers get the expected data either:

- from one of our standard outputs:
  - Vi550 communication module which exports inspection/review results with respect of the IPC CAMX standard.
  - The output file that Vision software can generate (text file)
- an output developed by Vi TECHNOLOGY and adapted to the customer needs. Customer must provide detailed specifications.

3 What’s new in version 5.0?

- JOINT_BRIDGE table replaced by new PIN and PIN_MEASURE tables. Data are not grouped in the same way:
  - In the previous JOINT_BRIDGE table, they were grouped by defect type (1 row could concern several pins)
  - In the PIN table, they are grouped by pin (1 row could concern 0, 1 or several defects).

PIN TABLE

Pin ID	Pin Identification				ErrorTable (Bit field)	ErrorTable_AR (Bit field)	Review Sanction	Review Comment	Bridge ID
	Tested Object ID	Component Side	Pin Index On Side	IPC Pin Nb					

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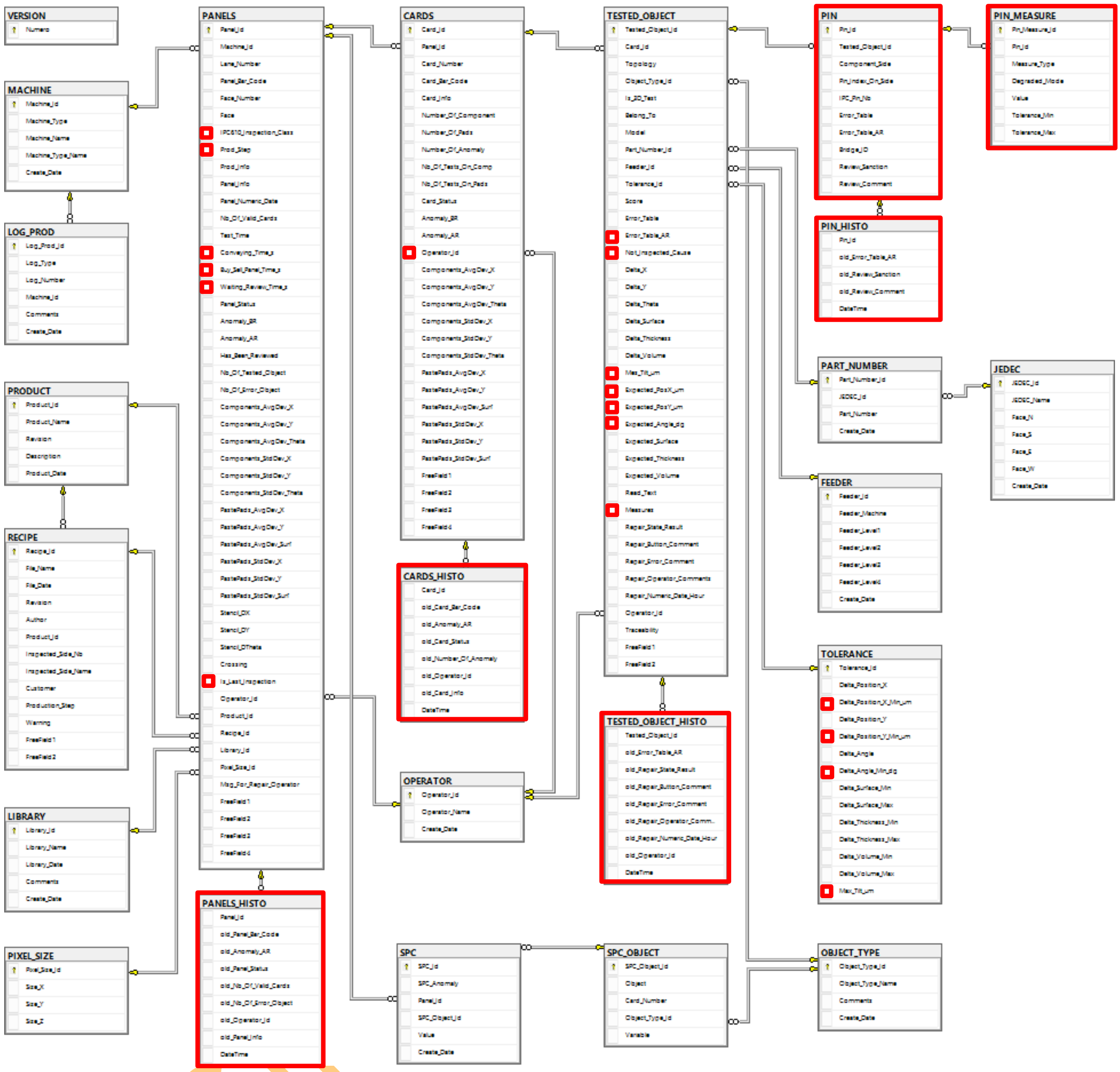
PIN_MEASURE TABLE

Pin ID	Measure Type	Value	Tolerance Min	Tolerance Max	Degraded Mode

- New tables for historization of data that can be changed after review.
  - PANELS_HISTO
  - CARDS_HISTO
  - TESTED_OBJECT_HISTO
  - PIN_HISTO
- New columns in tables
  - PANELS
  - CARDS
  - TESTED_OBJECT
  - TOLERANCE
- If we inspect several times a same panel side on the same machine, the previous inspection results are not deleted anymore.

4 Database structure



Legend: new tables and columns are marked in red.

The database is organized as a tree.
It starts with the table MACHINE: it can be an AOI or a review station.
In PANELS, you can find the list of panel inspection results of all the panels that has been inspected on a given machine. The tables are linked by Ids (auto-incremental values).
In CARDS, you can find the inspection results of all the sub-panels of a given panel.
In TESTED_OBJET, you can find the results of all the components (and solder paste…) inspected on a given sub-panel and the Foreign Materials discovered.
In PIN, you can find the status of all component pins (it gives the list of defects like a bad solder joint or a solder bridge between 2 pins).
In PIN_MEASURE, you can find the values of the measures made on the pin with the associated tolerances.

PANELS_HISTO, CARDS_HISTO, TESTED_OBJECT_HISTO, PIN_HISTO store the previous values of columns that are changed after a review in the associated tables.

The other tables give more information about the panels, their components, the inspection parameters...

5 Tables

5.1 Table PANELS

This table does not really present panels but panel inspection results:
1 record = 1 inspection of 1 side of 1 panel. A same side of a same panel can be inspected several times, at different production step or at the same production step in case of inspection issue.
A panel is identified by its code (barcode or DATAMATRIX code).
A side is identified by the associated side number (column Face_Number)
No Unique Constraint.
If we inspect several times a same panel side on the same machine, the column Crossing is incremented.
Remark: a panel can be referenced in the database while it has not been inspected. When a panel is ejected from the AOI without being inspected, we need to explain to the operator what it happened when the board arrives on a review station. So, we save it in the database.

Panel_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Panel_Bar_Code:	This column identifies a Panel. Generally, it corresponds to a barcode or DATAMATRIX code attached to the panel. If there is no barcode reader on the machine, this code is generated by the AOI with the inspection date-time.
Face_Number:	Identifies the side which has been inspected (0 or 1). -1 indicates that the side number is undefined in the used inspection program and that no dual side ID code reader has been used to determine it. Note that the process engineer should define the side number in the inspection program when the system is not able to deduce it.
Face:	Name associated to a side. It can be “Top”, “Bottom” …
Machine_Id:	Link to the “Machine” table. It identifies the AOI that inspected the panel
Lane_Number:	For DualLane AOI, it gives the internal lane number (1 or 2) where the panel has been inspected. 0 for single lane machine.
Prod_Step:	Production step: 0: undefined 1: pre-reflow 2: post-reflow
Prod_Info:	This column gives the lot number associated to the inspected panels (but the user can use it to store any kind of information that he needs to attach to the results of all panels inspected within the production session).
Panel_Info:	provides the operator comment in case of a global sanction as scrapped panel.
Panel_Numeric_Date:	Date and time of the inspection. It is an ANSI time_t value. It is the number of seconds elapsed since midnight (00:00:00), January 1, 1970, coordinated universal time, according to the system clock.

Conveying_Time_s: Time (in seconds) to convey the panel inside and outside the AOI. NOT AVAILABLE YET.

Test_Time: inspection duration (in s). It does not include the time necessary to convey the panel inside and outside the AOI.

Buy_Sell_Panel_Time_s: Time (in seconds) to wait upstream and downstream machine. NOT AVAILABLE YET.

Waiting_Review_Time_s: Time (in seconds) to wait the end of the review (for embedded review only). NOT AVAILABLE YET.

Product_Id: Link to the Product table. It identifies the inspection program.

Library_Id: Link to the Library table. It identifies the inspection model library used with the inspection program.

IPC610_Inspection_Class: from class 1 to 3. 0 when there is no IPC 610 test.

Nb_Of_Valid_Cards: Number of good sub-panels.

Nb_Of_Tested_Object: gives the number of components and paste pads that has been inspected on this panel.

Nb_Of_Error_Object: gives the number of faulty components and paste pads. This value is updated after reviewing. This value does not include defects of invalid sub-panels. This value equals 0 if the operator has washed the panel on the review station.

Has_Been_Reviewed: Indicates that the panel has been presented on a review station.
0: not passed on a review station
1: passed on a review station (even if the operator pressed on Escape)
255: transitory state that indicates that the panel results are not fully saved. It is used by offline FIFO Review to not get results before they are ready.

Anomaly_BR (before reviewing) and Anomaly_AR (after reviewing): It gives the global state of the panel. It's a field of 12 bits. You can use a mask to have the state of the panel. If there's no error, this field equals 0.
After inspection and before reviewing, Anomaly_BR = Anomaly_AR.

	Binary	decimal	meaning
bit 1	0000 0000 0001	1	fiducial error -> the panel has not been inspected
bit 2	0000 0000 0010	2	code reading error (barcode or DATAMATRIX code)
bit 3	0000 0000 0100	4	unexpected code root (barcode or DATAMATRIX code)
bit 4	0000 0000 1000	8	This panel has been ejected by the review operator
bit 5	0000 0001 0000	16	This panel has been washed by the review operator
bit 6	0000 0010 0000	32	found one or more defects on the panel (see faulty components and faulty paste pads in the table TESTED_OBJECT).
bit 7	0000 0100 0000	64	not specified error. In General, it is used by communication modules when a panel is ejected without being inspected.
Bit 8	0000 1000 0000	128	axis error (if an axis error happens, the panel results should not be saved in the database).
bit 9	0001 0000 0000	256	This panel has not been inspected. All its sub-panels have been skipped

bit 10	0010 0000 0000	512	One of its sub-panels has an error on its code (barcode or DATAMATRIX code)
bit 11	0100 0000 0000	1024	There are too many defects on the panel (the limit has been reached), so they have not been saved in the table TESTED_OBJECT.
bit 12	1000 0000 0000	2048	This is the reference to know if the panel has been inspected (=0) or not (=1). Panels are saved in the database while they have not been inspected in order to present the problem to the operator when these panels arrive on a review station.

Panel_Status: gives the panel status. It is like Anomaly_AR but it ignores the ID code reading error because this is not a panel defect and it is not a binary field. That way, it can be used as a filter in database requests.

Type: tiny int

Value	Meaning
-2	Still faulty after reviewing
-1	Faulty (after inspection)
0	not inspected
1	Good (after inspection)
2	Good because all defects were dummy faults
3	Good after reviewing

Msg_For_Repair_Operator: It is used for panels that have been ejected from the AOI without being inspected. When the panel arrives on a review station, this message is shown to the operator. That way, the operator knows that the panel has not been inspected and why.

Crossing: Number of times that this panel side has been inspected on the same machine.

Is_Last_Inspection: Indicates if this row corresponds to the last inspection of this side of this panel on this machine: 0 = FALSE, 1 = TRUE.

Operator_Id: Link to the Operator table that gives the review operator name. It is used in case of a global sanction as scrapped panel.

Pixel_Size_Id: Link to the Pixel_Size table. Obsolete column. It was used by old generation of 3D inspection machines.

Columns used to calculate the Cp values:

Components_AvgDev_X: average deviation of components on X axis (µm). Type: double.

Components_AvgDev_Y : average deviation of components on Y axis (µm). Type: double.

Components_AvgDev_Theta: average rotation of components (degree). Type: double.

Components_StdDev_X: standard deviation of components on X axis (µm). Type: double.

Components_StdDev_Y: standard deviation of components on Y axis (µm). Type: double.

Components_StdDev_Theta: standard rotation of components (degree). Type: double.

PastePads_AvgDev_X: average deviation of paste pads on X axis (µm). Type: double.

PastePads_AvgDev_Y: average deviation of paste pads on Y axis (µm). Type: double.

PastePads_AvgDev_Surf: average of paste pad surface area errors (μm^2). Type: double.

PastePads_StdDev_X: standard deviation of paste pads on X axis (µm). Type: double.
PastePads_StdDev_Y: standard deviation of paste pads on Y axis (µm). Type: double.
PastePads_StdDev_Surf: standard deviation of paste pad surface area errors (μm^2). Type: double.
Stencil_DX, Stencil_DY, Stencil_DTheta: when the AOI is just after a printer machine, we can calculate the deviation of the stencil using the deviation of all the inspected paste pads. Units: micron and degree.
FreeFields (1->4): columns used for custom purpose.

5.2 Table CARDS (sub-panels)

This table gives information about sub-panels.
On a panel, we identify all its sub-panels by their associated number (see the column Card_Number). These numbers come from the inspection program (TST file).
Unique Constraint: combination of the columns Panel_Id and Card_Bar_Code.

Card_Id: Auto incremental value generated by the database engine. It is the Primary Key of this table. This is a BIGINT since version 5.0 of the database.

Panel_Id: Link to the panel to which this sub-panel belongs.

Card_Bar_Code: If the sub-panel has its own code (barcode or DATAMATRIX code), it is saved in this column. It can be used to identify the sub-panel. It is often generated by the AOI using the panel code.

Card_Number: Number affected to the sub-panel (1 to n).

Card_Info: provides the operator comment in case of a global sanction as scrapped sub-panel.

Anomaly_BR (before reviewing) and Anomaly_AR (after reviewing): It gives the global state of one sub-panel of the panel. It's a field of 12 bits. You can use a mask to have the state of the sub-panel. If there's no error, this field equals 0.

After inspection and before reviewing, Anomaly_BR = Anomaly_AR.

	Binary	decimal	meaning
bit 1	0000 0000 0001	1	Fiducial error -> the panel has not been inspected
bit 2	0000 0000 0010	2	Code reading error (barcode or DATAMATRIX code)
bit 3	0000 0000 0100	4	Unexpected code root (barcode or DATAMATRIX code)
bit 4	0000 0000 1000	8	This panel has been ejected by the review operator
bit 5	0000 0001 0000	16	This panel has been washed by the review operator
bit 6	0000 0010 0000	32	Found one or more defects on this sub-panel (see faulty components and faulty paste pads in the table TESTED_OBJECT).
bit 7	0000 0100 0000	64	Not specified error. In General, it is used by communication modules when a panel is ejected without being inspected.
bit 8	0000 1000 0000	128	Axis error (if an axis error happens, the panel results should not be saved in the database).
bit 9	0001 0000 0000	256	skipped sub-panel
bit 10	0010 0000 0000	512	This sub-panel has been invalidated by the review operator. The panel status does not take care of its invalidated sub-panels. The panel can be considered as a good panel. If all the sub-panels are invalidated, the panel is considered as ejected (Panel. Anomaly_AR = 8)
bit 11	0100 0000 0000	1024	There are too many defects on the panel (the limit has been reached), so they have not been saved in the table TESTED_OBJECT.

bit 12	1000 0000 0000	2048	This is the reference to know if this sub-panel has been inspected (=0) or not (=1).

Card_Status: gives the sub-panel status. It is similar to Anomaly_AR but it ignores the ID code reading error because this is not a panel defect and it is not a binary field. That way, it can be used as a filter in database requests.

Type: tiny int

Value	Meaning
-2	Still faulty after reviewing
-1	Faulty (after inspection)
0	not inspected
1	Good (after inspection)
2	Good because all defects were dummy faults
3	Good after reviewing

Operator_Id: Link to the Operator table that gives the review operator name. It is used in case of a global sanction as "scrapped sub-panel."

Number_Of_Component: Number of components inspected on this sub-panel.

Number_Of_Pads: Number of paste pads inspected on this sub-panel.

Number_Of_Anomaly: Number of faulty components and paste pads found on this sub-panel.

This column is like the column in Panels.Nb_Of_Error_Object. However, it is not updated in the same way after reviewing. If the panel is washed or if the sub-panel is invalidated by the review operator, this column keeps its initial value. Otherwise, it is updated accordingly to the operator actions.

Nb_Of_Tests_On_Comp: Number of unitary tests performed to inspect the components of the sub-panel. It is used to calculate the DPMO. Type: long integer.

Nb_Of_Tests_On_Pads: Number of unitary tests performed to inspect the paste pads of the sub-panel. It is used to calculate the DPMO. Type: long integer.

Fields used to calculate the Cp values:

Components_AvgDev_X: average deviation of components on X axis (µm). Type: double.

Components_AvgDev_Y: average deviation of components on Y axis (µm). Type: double.

Components_AvgDev_Theta: average rotation of components (degree). Type: double.

Components_StdDev_X: standard deviation of components on X axis (µm). Type: double.

Components_StdDev_Y: standard deviation of components on Y axis (µm). Type: double.

Components_StdDev_Theta: standard rotation of components (degree). Type: double.

PastePads_AvgDev_X: average deviation of paste pads on X axis (µm). Type: double.

PastePads_AvgDev_Y: average deviation of paste pads on Y axis (µm). Type: double.

PastePads_AvgDev_Surf: average of paste pad surface area errors (µm²). Type: double.

PastePads_StdDev_X: standard deviation of paste pads on X axis (µm). Type: double.

PastePads_StdDev_Y: standard deviation of paste pads on Y axis (µm). Type: double.

PastePads_StdDev_Surf: standard deviation of paste pad surface area errors (µm²). Type: double.

FreeFields (1->4): columns used for custom purpose.

5.3 Table TESTED_OBJECT (components, texts alone, macros, foreign materials)

It gives the results of a tested object. It can be a component, a paste pad, a connector, a pin of a connector, a text (OCV tests are often associated to a component, they are not alone).

But it can also provide:

- the result of a macro (distance or angle between 2 components, or with a given line…),
- the position and the encompassing area of a detected foreign material.

Generally, only the faulty objects are saved in the database (for performance reason). But it is also possible to save all the tested objects.

Unique Constraint: combination of the columns Card_Id and Topology.

Tested_Object_Id: Auto incremental value generated by the database engine. It is the Primary Key of this table. This is a BIGINT since version 5.0 of the database.

Card_Id: Link to the sub-panel to which this TESTED_OBJECT belongs. This is a BIGINT since version 5.0 of the database.

Topology: Name of the component given by the customer (= reference designator). It is unique within a sub-panel.

Object_Type_Id: Link to the Object_Type table that gives the kind of the tested object: Component, paste pad…

Part_Number_Id: Link to the Part_Number table. It is the Manufacturer reference of the component. To apply to "Components" or "Connectors". In the case of a macro, it gives the kind of macro.

Belong_To: Obsolete column. To apply to "Pins" of "Connectors". It gives the topology of the connector to which it belongs.

Inspection Results:

Error_Table: It gives the list of defects found on this tested object by the AOI (provided in a field of bits). It is not updated after the operator review. Looking if this field equals 0 is not enough to know if this component is good, it is also necessary to check that it has been inspected with the Not_Inspected_Cause field

	Binary	decimal	meaning
bit 1	0000 0000 0000 0000 0000 0000 0001	1	object missing
bit 2	0000 0000 0000 0000 0000 0000 0010	2	polarity error
bit 3	0000 0000 0000 0000 0000 0000 0100	4	solder joint defect (refer to PIN table)
bit 4	0000 0000 0000 0000 0000 0000 1000	8	solder bridge defect (refer to PIN table)
bit 5	0000 0000 0000 0000 0000 0001 0000	16	OCV error
bit 6	0000 0000 0000 0000 0000 0010 0000	32	model not found in library (NOT used anymore, replaced by the column Not_Inspected_Cause)
bit 7	0000 0000 0000 0000 0000 0100 0000	64	Delta_X value out of range
bit 8	0000 0000 0000 0000 0000 1000 0000	128	Delta_Y value out of range
bit 9	0000 0000 0000 0000 0001 0000 0000	256	Delta_Theta value out of range
bit 10	0000 0000 0000 0000 0010 0000 0000	512	Delta_Thickness out of range
bit 11	0000 0000 0000 0000 0100 0000 0000	1024	paste surface area out of range

bit 12	0000 0000 0000 0000 1000 0000 0000	2048	element skipped (NOT used anymore, replaced by the column Not_Inspected_Cause)
bit 13	0000 0000 0000 0001 0000 0000 0000	4096	Connector: the space between 2 pin columns is not respected (obsolete)
bit 14	0000 0000 0000 0010 0000 0000 0000	8192	Connector: the space between 2 pin rows is not respected (obsolete)
bit 15	0000 0000 0000 0100 0000 0000 0000	16384	Connector: a pin is missing (obsolete)
bit 16	0000 0000 0000 1000 0000 0000 0000	32768	Connector: bad pin alignment (obsolete)
bit 17	0000 0000 0001 0000 0000 0000 0000	65536	Volume value out of range (obsolete)
bit 18	0000 0000 0010 0000 0000 0000 0000	131072	Bad appearance (obsolete)
bit 19	0000 0000 0100 0000 0000 0000 0000	262144	Potential defect imported from SPI
bit 20	0000 0000 1000 0000 0000 0000 0000	524288	Tilt error (bad coplanarity)
bit 21	0000 0001 0000 0000 0000 0000 0000	1048576	Side overhang (IPC 610)
bit 22	0000 0010 0000 0000 0000 0000 0000	2097152	Length overhang (IPC 610)
bit 23	0000 0100 0000 0000 0000 0000 0000	4194304	Foreign material detected
Bit 24	0000 1000 0000 0000 0000 0000 0000	8388608	Component present (it should not)
Bit 25	0001 0000 0000 0000 0000 0000 0000	16777216	Lifted lead (refer to PIN Table)

Error_Table_AR: It gives the **current list** of defects found of this tested object. This is the same as Error_Table but it is updated after the operator review by the classification (the operator can add or remove or change some defects) and by the sanction (ex: a dummy fault sanction removes all the defects). If the component has no defect, this field has the value 0. This is a BIGINT, its 32 upper bits are used by classification.
Looking if this field equals 0 is not enough to know if this component is good, it is also necessary to check that it has been inspected with the Not_Inspected_Cause field.

Not_Inspected_Cause: Indicates if this component has been inspected or not and why:

0	Inspected
1	ManuallySkipped: the operator chose to skip the inspection of this component. In previous version of the database, this information was provided using a specific bit in the Error_table column
2	ModelNotFound
3	BadProgramming

Score: Score of the winner model.

Delta_X: Deviation of the component on the X axis (µm).
In the case of a macro, this field provides the calculated distance (it can be a distance on the x axis or not, it depends on the macro type provided by the associated Part Number).
In case of a ForeignMaterial, this field provides the X position of its encompassing area center.

Delta_Y: Deviation of the component on the Y axis (µm).
In the case of a macro, this field provides the calculated distance (it depends on the macro type provided by the associated Part Number).
In case of a ForeignMaterial, this field provides the Y position of its encompassing area center.

Delta_Theta: Rotational deviation of the component (degree) In the case of a macro, this field provides the calculated angle (it depends on the macro type provided by the associated Part Number).

Delta_Thickness: Difference between real and expected thickness (µm). In case of a ForeignMaterial, this field provides its height.

Delta_Surface: Difference between real and expected surfaces (%)

Delta_Volume: Obsolete column. Difference between real and expected volumes (%)

Mes_Tilt_um: Difference of height between the lowest and the highest points of the treatment area (µm)

Expected_PosX_um: Used for macros only, 0 in the other cases (µm)

Expected_PosY_um: Used for macros only, 0 in the other cases (µm)

Expected_Angle_dg: Used for macros only, 0 in the other cases (degree)

Expected_Thickness: Expected thickness (µm). In case of a ForeignMaterial, this field provides the height in µm of its encompassing area.

Expected_Surface: Expected surface (μm^2)

Expected_Volume: Expected volume (μm^3). In case of a ForeignMaterial, this field provides the width in µm of its encompassing area.

Read_Text: text read by OCR treatment or text checked by OCV or text read by a Trace object (‘ID code’ model type result).

Measures: Values of the internal treatment variables specified in the ‘Measures’ equation of the winner model.

Inspection conditions:

Model: Name of the model used to inspect the component. This column is used by the Reparation software. To apply to “Components”.

Tolerance_Id: Link to the Tolerance table. It gives the test tolerances.

Is_3D_Test: 0 or 1 if the object has been inspected with a 3D test (at least one of its models does a 3 test, even if the winner model doesn’t use it).

Review result and information:

Repair_State_result: It gives the current state of the component: values <= 0 are given by the AOI, values > 0 are given by the Review.

-2	not inspected (manually skipped or bad programming)
-1	not detected as faulty by AOI
0	detected as faulty by AOI but not reviewed yet
1	repaired
2	good (acceptable or dummy fault)
3	confirmed as faulty

Repair_Button_Comment: Name of the button selected by the operator

Repair_Error_Comment: Comment associated to the button selected by the operator

Repair_Operator_Comment: Operator comment

Repair_Numeric_Date_Hour: Date and time of the operator’s diagnosis. It is an ANSI time_t value. It is the number of seconds elapsed since midnight (00:00:00), January 1, 1970, coordinated universal time, according to the system clock.

Operator_Id: Link to the Operator table that gives the review operator name.

Miscellaneous:

Feeder_Id:	Link to the Feeder table. It identifies the placement machine, the gantry, the feeder, and the nozzle used to place the component. Used with “Components” or “Connectors”.
Traceability:	0 or 1 to indicate that the user wants to keep the inspection results of this component even when it is good.
FreeFields (1->2):	columns used for custom purpose.

5.4 Table PIN

This table provides the status of all component pins (it gives the list of pin defects like a bad solder joint or a solder bridge between 2 pins).
1 row = 1 pin.

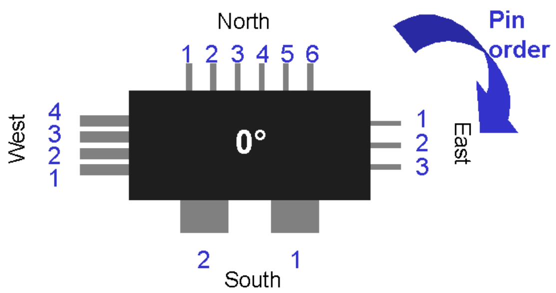
Tested_Object_Id: Link to the TESTED_OBJECT to which this pin belongs.

Pin_Id: Auto incremental value generated by the database engine. It is the Primary Key of this table.

Component_Side: Indicates on which component side the pin belongs to. It is relative to the component at 0 degree as it is defined in the library

0	top side (North)
1	right side (East)
2	bottom side (South)
3	left side (West)

Pin_Index_On_Side: Index of the pin on its component side
They are indexed on 0 (0 corresponds to the 1st pin of the side).
They are increased following the clockwise direction.



IPC_Pin_Nb: Global number of this pin as specified in IPC standard.
0 indicates that is value is not defined.

Error_Table It gives the list of defects found on this pin by the AOI. It is not updated after the operator review. As TESTED_OBJECT.Error_Table, this a binary field. Each bit corresponds to a kind of defect (it uses the same bits as TESTED_OBJECT.Error_Table). If there's no defect, this field has the value 0.

Error_Table_AR It gives the **current list** of defects found on this pin. This is the same as Error_Table but it is updated after the operator review by the classification (the operator can add or remove or change some defects) and by the sanction (ex: a dummy fault sanction removes all the defects)

Bridge_Id This value identifies each bridge found on a panel by a unique number (unique for a panel inspection). It allows to group all pins that belongs to a same bridge.
0 when this pin doesn't belong to a bridge.

Review_Sanction provides the last operator sanction on this pin.

-1	not in error (can happen only in case of full traceability)
0	detected as faulty by AOI but not reviewed yet
1	repaired
2	good (acceptable or dummy fault)
3	confirmed as faulty

Review_Comment Comment of the operator who made the sanction.

5.5 Table PIN_MEASURE

This table provides the measures made on a pin with the associated tolerance values.

Pin_Id: Link to the PIN on which this measure was made.

Pin_Measure_Id: Auto incremental value generated by the database engine. It is the Primary Key of this table.

Measure_Type: Kind of measure:

1	SideOverhang_per
2	LengthOverhang_um
3	Joint3D_Height_um
4	Joint3D_Width_um
5	Joint3D_Length_um
6	Joint3D_Convexity
7	Joint3D_TailVolume_mm3
8	Joint3D_Volume_mm3
9	Joint3D_Uniformity_per
10	Joint2D_Average_g
11	Joint2D_Area_per
12	LeadHeight_um
13	Joint3D_TerminationHeight_um
14	Joint3D_WettingAngle_deg
15	Joint3D_Height_per
16	Joint3D_Width_per
17	Joint3D_TailVolume_per

Value: Measured value. Its unit depends on the kind of measure:
- SideOverhang values are %
- LengthOverhang values are in µm
- Joint3D_Height values are in µm
- Joint3D_Width values are in µm
- Joint3D_Length values are in µm
- Joint3D_Convexity (-1->1)
- Joint3D_TailVolume values are in mm3
- Joint3D_Volume values are in mm3
- Joint3D_Uniformity values are %
- Joint2D_Average values are grey level (0->255)
- Joint2D_Area values are %
- LeadHeight values are in µm
- Joint3D_TerminationHeight values are in µm
- Joint3D_WettingAngle values are in degrees
- Joint3D_Height values are %
- Joint3D_Width values are %
- Joint3D_TailVolume values are %

Tolerance_Min: Associated tolerance.

Tolerance_Max: Associated tolerance.

Degraded_Mode: Indicates that the measure has been done on a 2D image in place of 3D (it can happen when the 3D image quality is not good enough at this position). At- present it is a boolean value (0/1) but it could be extended in the future with more values for different cases.

5.6 Table PANELS_HISTO

Stores the previous values of PANELS columns that have been changed following an operator review.
As soon as any of the following columns are modified in the PANELS table, a row is inserted in this table to save the previous values. Note that some of them are copied while they were not changed.

Panel_Id:	ID of the PANELS table row that has been modified.
old_Panel_Bar_Code:	Previous value.
old_Anomaly_AR:	Previous value.
old_Panel_Status:	Previous value.
old_Nb_Of_Valid_Cards:	Previous value.
old_Nb_Of_Error_Object:	Previous value.
old_Operator_Id:	Previous value.
old_Panel_Info:	Previous review operator comment.
DateTime:	UTC Date of the modification.

5.7 Table CARDS_HISTO

Stores the previous values of CARDS columns that have been changed following an operator review.
As soon as any of the following columns are modified in the CARDS table, a row is inserted in this table to save the previous values. Note that some of them are copied while they were not changed.

Card_Id:	ID of the CARDS table row that has been modified.
old_Card_Bar_Code:	Previous value.
old_Anomaly_AR:	Previous value.
old_Card_Status:	Previous value.
old_Number_Of_Anomaly:	Previous value.
old_Operator_Id:	Previous value.
old_Card_Info:	Previous review operator comment.
DateTime:	UTC Date of the modification.

5.8 Table TESTED_OBJECT_HISTO

Stores the previous values of TESTED_OBJECT columns that have been changed following an operator review.
As soon as any of the following columns are modified in the TESTED_OBJECT table, a row is inserted in this table to save the previous values. Note that some of them are copied while they were not changed.

Tested_Object_Id:	ID of the TESTED_OBJECT table row that has been modified.
old_Error_Table_AR:	Previous value.
old_Repair_State_Result:	Previous value.

old_Repair_Button_Comment: Previous value.
old_Repair_Error_Comment: Previous value.
old_Repair_Operator_Comments: Previous value.
old_Repair_Numeric_Date_Hour: Previous value.
old_Operator_Id: Previous value.
DateTime: UTC Date of the modification.

5.9 Table PIN_HISTO

Stores the previous values of PIN columns that have been changed following an operator review.
As soon as any of the following columns are modified in the PIN table, a row is inserted in this table to save the previous values. Note that some of them are copied while they were not changed.

Pin_Id:	ID of the PIN table row that has been modified.
old_Error_Table_AR:	Previous value.
old_Review_Sanction:	Previous value.
old_Review_Comment:	Previous value.
DateTime:	UTC Date of the modification.

5.10 Table MACHINE

This table lists the machines that are connected (through a Supervisor) to this database.

Machine_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Machine_Name:	Name of the machine (name of its computer)
Machine_Type:	1 = AOI 2 = Review station
Machine_Type_Name:	same as Machine_Type but readable form.
Create_Date:	Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.11 Table PRODUCT

This table lists the products (and their revisions) that has been inspected by the AOIs.
Unique Constraint: combination of the columns Product_Name and Revision.

Product_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Product_Name:	Product name (defined in the TST property page)
Revision:	Revision of the product (defined in the TST property page)
Description:	Product description (defined in the TST property page). If this column is modified but not the revision column, the previous description is erased (we keep the description provided by the last used TST).
Product_Date:	It corresponds to the record creation date.

5.12 Table RECIPE

This table contains all inspection programs (and their revisions) that has been used by the AOIs.
Different inspection programs can correspond to a given product:

- The inspection program can be linked to a face: top or bottom or Both in case of symmetrical panels
- We can have different versions of an inspection program depending on the placement of the AOI in the production line. An inspection program can be associated to a production step (before or after oven…)
- Evolution of an inspection program -> X versions of the same inspection program: we differentiate the inspection program using the last TST modification date.

When the date of the TST file (= inspection program) changes, we consider the inspection program has been changed so we save it again with a new ID.

Unique Constraint: combination of the columns File_Name and File_Date.

Recipe_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
File_Name:	Inspection program file name (without path and extension)
File_Date:	Date of the inspection program file (last modification). It is an ANSI time_t value. It is the number of seconds elapsed since midnight (00:00:00), January 1, 1970, coordinated universal time, according to the system clock.
Product_ID:	Link to the Product table. It identifies the associated product.
Inspected_Side_Nb:	number of the board side inspected with this recipe (defined in TST property page) -1: this recipe can be used to inspect both sides. This is the case of symmetrical boards. 0: it can be Top or Bottom side, it depends on the customer strategy. 1: it can be Top or Bottom side, it depends on the customer strategy.
Inspected_Side_Name:	name of the board side inspected with this recipe (defined in TST property page)
Author:	Author of the inspection program (defined in TST property page)
Revision:	Revision of the inspection program (defined in TST property page)
Production_Step:	Production step where this recipe should be used (defined in TST property page)
Customer:	customer for whom the product is dedicated (defined in TST property page)
FreeFields (1->2):	columns used for custom purpose.

5.13 Table LIBRARY

A library defines how to inspect each kind of components. An inspection program specifies which test of the library (=model) should be used for each component of the panel. We do not need a library when we only inspect paste pads.
When the library file date changes, we consider the library has been changed so we save it again with a new ID.
Unique Constraint: combination of the columns Library_Name and Library_Date.

Library_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Library_Name:	Test library file name (without path and extension)
Library_Date:	Date of the library file. It is an ANSI time_t value. It is the number of seconds elapsed since midnight (00:00:00), January 1, 1970, coordinated universal time, according to the system clock.
Comments:	for database administrator
Create_Date:	Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.14 Table OPERATOR

This table lists the operator's names. Currently, it only includes review operator's names.

Operator_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Operator_Name:	Operator's name.
Create_Date:	Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.15 Table PIXEL_SIZE (obsolete)

This table gives the pixel size that is used on the AOI that inspected the panel. It was used by old generation of 3D inspection machines.

Pixel_Size_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Size_X, SizeY:	Width and Length of pixels (micron)
Size_Z:	Height of a gray level (micron)

5.16 Table TOLERANCE

This table saves the tolerances used for each inspected object (component or paste pad).
One record gives all the tolerances used to inspect an object.
Following the kind of object, only some tolerances are used.
Unique Constraint: combination of all the tolerance values.

Tolerance_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Delta_Position_X_Min_um:	Minimum deviation accepted on the X axis (µm). Mainly required for macros because in the other cases, this value is the opposite of max value.
Delta_Position_X:	Maximum deviation accepted on the X axis (µm)
Delta_Position_Y_Min_um:	Minimum deviation accepted on the Y axis (µm). Mainly required for macros because in the other cases, this value is the opposite of max value.
Delta_Position_Y:	Maximum deviation accepted on the Y axis (µm)
Delta_Angle_Min_dg:	Minimum rotational deviation accepted (degree). Mainly required for macros because in the other cases, this value is the opposite of max value.
Delta_Angle:	Maximum rotational deviation accepted (degree). It is used only for components and macros.
Delta_Thickness_Min:	Maximum difference between real and expected.
Delta_Thickness_Max:	Maximum difference between real and expected thickness.
Delta_Surface_Min:	Maximum difference between real and expected surfaces if there is not enough paste (%). It is used only for paste pads.
Delta_Surface_Max:	Maximum difference between real and expected surfaces if there is too many paste (%). It is used only for paste pads.
Delta_Volume_Min:	Obsolete column. Maximum difference between real and expected volumes if there is not enough paste (%). It is used only for paste pads.
Delta_Volume_Max:	Obsolete column. Maximum difference between real and expected volumes if there is too many paste (%). It is used only for paste pads.
Max_Tilt_um:	Maximum tilt value (µm)

5.17Table PART_NUMBER

A Part-Number is the Manufacturer reference of a component.
This table gives the part-number of each inspected component.
It can also contain the types of the used macros:

- DISTANCE_X
- DISTANCE_Y
- DISTANCE
- ANGLE_2_TOPO
- ANGLE_STRAIGHTLINE_TOPO
- DISTANCE_POS+ANGLE_STRAIGHTLINE_2_TOPO
- ANGLE_3_TOPO
- DISTANCE_2_TOPO_STRAIGHTLINE_TOPO
- DISTANCE_TOPO1_TO_PROJECTED_TOPO3_ON_STRAIGHTLINE_TOPO1_TOPO2

Part_Number_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Part_Number:	reference of the component
Jedec_Id:	Link to the JEDEC table. It identifies the shape of the component.
Create_Date:	Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.18Table JEDEC

A JEDEC identifies a component shape. This table gives the number of pins on each side of a component.
It can also contain the types of the used macros:

- DISTANCE_X
- DISTANCE_Y
- DISTANCE
- ANGLE_2_TOPO
- ANGLE_STRAIGHTLINE_TOPO
- DISTANCE_POS+ANGLE_STRAIGHTLINE_2_TOPO
- ANGLE_3_TOPO
- DISTANCE_2_TOPO_STRAIGHTLINE_TOPO
- DISTANCE_TOPO1_TO_PROJECTED_TOPO3_ON_STRAIGHTLINE_TOPO1_TOPO2

Jedec_Id:	Auto incremental value generated by the database engine. It is the Primary Key of this table.
Jedec_Name:	Name that identifies a component shape
Face_N:	number of pins on the North side of the component
Face_S:	number of pins on the Sud side of the component
Face_E:	number of pins on the Est side of the component
Face_W:	number of pins on the West side of the component
Create_Date:	Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.19 Table FEEDER

This table gives information about placement machines. These information are imported at one time before launching a production session. Afterwards, they are not updated so it is not possible to verify their validity.
Generally, this table is populated with placement machine name, nozzle gantry and feeder that have been used to place a component. But it depends of the imported files.
Unique Constraint: combination of all the columns (except Feeder_Id)

Feeder_Id: Auto incremental value generated by the database engine. It is the Primary Key of this table.
Feeder_Machine: Placement machine name
Feeder_ Level_1->4: Identify placement machine parts
Create_Date: Creation date of this record (for database administrator). It is a float value, but it is readable. Example: 2.0050128183733 -> 2005/01/28 18:37:33

5.20 Table OBJECT_TYPE

This table presents the different kinds of tested objects: component, paste pad…
It is designed for the database administrator.
Object_Type_Id: code which identifies the kind of the tested object

hexadecimal	decimal	meaning
0x00000001	1	Component
0x00000008	8	Text (text alone, not associated to a component)
0x00000010	16	Paste pad
0x00001000	4096	Macro (combination of several tests)
0x00010000	65536	Connector (obsolete)
0x00020000	131072	Group of connectors (obsolete)
0x02000000	33554432	FOREIGN_MATERIAL

Object_Type_Name: readable form of the code
Comments: for database administrator.

5.21 Table SPC (obsolete)

This table saves the SPC warnings generated during the production by the embedded SPC engine.
A warning does not concern only one panel but a group of panels. In the database, a warning is attached to the last panel that make it occurred.

5.22 Table SPC_Object (obsolete)

This table identifies the SPC variables that generated a warning.
SPC variables: it can be one of the inspection results (like Delta_X) of specific component which we are studying the variations (Example: the Delta_X values of the Component C001)

5.23 Table LOG_PROD

This table is not used yet. It should store the machine working transitions like “load new board”, “start inspection”, “complete inspection” …

6 Records by default

The following tables have a record by default with the ID 1:

- LIBRARY,
- OPERATOR,
- TOLERANCE
- PART_NUMBER
- JEDEC,
- FEEDER.

They are used for the database integrity.

Example: when we do not have the placement machine data, the TESTED_OBJECTs point to the Feeder_Id 1.

7 Maintenance

When the database size becomes huge, it is necessary to either:

- delete the oldest results in order to reduce its size and not overload the database server,
- or continue the production on a new database but it brings the issue to lose the most recent results (almost those of panels that have not been reviewed yet). So, it is necessary to import the most recent results in the new database.

Vi TECHNOLOGY doesn't provide any tool to do that, so it is necessary to do that manually with some SQL requests.

In both solutions, keep in minds that tables are linked with each other's with foreign keys: TESTED_OBJECT rows have a Card_Id column which points to 1 row of the CARDS table, CARDS rows have a Panel_Id column which points to 1 row of the PANELS table… So, it is not possible to add a row in CARDS before creating the associated row in PANELS. For the same reason, it is not possible to delete a row in PANELS before deleting all the rows in CARDS which point to this row (this is not true anymore with SQL server databases created with Vision3D CR4 (or newest ones) because they are deleted in cascade).
The chart displayed in §4 au-dessus shows all the dependencies.

1st solution recommendations:
You just have to focus on PIN_MEASURE, PIN, TESTED_OBJECT, CARDS and PANELS tables and delete their rows in this order.
To select the panel results that you want to delete you can select those whose:

- Panel_Numeric_Date are under than your datetime limit
- Panel_Status is >= 0 if you want to delete all good panels
- HAS_BEEN_REVIEWED column is equal to 1 if you want to delete all reviewed panels

2nd solution recommendations:
For each panel that you want to copy in the new database:

- you must first insert the associated rows in the tables MACHINE, PRODUCT, RECIPE, LIBRARY, OPERATOR, PIXEL_SIZE.
- Then read the ID of the new rows and use them when you insert the new PANELS row (don't keep the original values of Recipe_Id, Product_Id… because there are specific to the original database).
- Before inserting the associated CARDS rows, you have to insert, for each of them, the associated rows in the OPERATOR table.
- Read the ID of each of them and also the PANEL_ID then insert the CARDS rows.
- Before inserting the associated TESTED_OBJECT rows, you have to insert, for each of them, the associated rows in the following tables: JEDEC, PART_NUMBER, TOLERANCE, OPERATOR and FEEDER.
- Read the ID of each of them and also the CARD_ID then insert the TESTED_OBJECT rows.
- Read the TestedObject_Id of the new TESTED_OBJECT rows and insert the associated rows in the PIN table.
- Read the Pin_Id of the new PIN rows and insert the associated rows in the PIN_MEASURE table.

To select the panel results that you want to copy, you can select those whose:

- Panel_Numeric_Date are upper than your datetime limit
- Panel_Status is < 0 if you want to keep only faulty panels
- HAS_BEEN_REVIEWED column is equal to 0 if you want to keep only not reviewed panels